14

ZERO–FREE REGIONS FOR $L(s, \chi)$

There is no difficulty in extending the results of the preceding section to the zeros of $L(s, \chi)$ when χ is a *fixed* character. But this is of limited value; for many purposes it is important to allow q to vary and to have estimates that are explicit in respect of q. This raises some difficult problems, and the results so far known are better for complex characters than for real characters.

We no longer suppose that $t \geq 2$ but merely that $t \geq 0$. There is no loss of generality in the latter supposition, for the zeros of $L(s, \chi)$ with $t < 0$ are the complex conjugates of the zeros of $L(s, \bar{\chi})$ with $t > 0$. We are concerned with nonprincipal characters only, and therefore $q \geq 3$ throughout.

Logarithmic differentiation of the Euler product formula gives

$$-\frac{L'(s, \chi)}{L(s, \chi)} = \sum_{n=1}^{\infty} \Lambda(n) n^{-\sigma} \chi(n) e^{-it \log n}$$

for $\sigma > 1$. We can represent the real part of $\chi(n)e^{-it \log n}$, for $(n, q) = 1$, as $\cos \theta$, and θ has to be replaced by 2θ if χ is replaced by χ^2 and t by $2t$, and has to be replaced by 0 if χ is replaced by χ_0 and t by 0. Hence the analog of the inequality (3) of the preceding section is

$$3\left[-\frac{L'(\sigma, \chi_0)}{L(\sigma, \chi_0)}\right] + 4\left[-\Re\frac{L'(\sigma + it, \chi)}{L(\sigma + it, \chi)}\right] + \left[-\Re\frac{L'(\sigma + 2it, \chi^2)}{L(\sigma + 2it, \chi^2)}\right] \geq 0. \tag{1}$$

If χ is a real character (but only then) we have $\chi^2 = \chi_0$, and this affects the argument. The effect is important only when t is small and we come under the influence of the pole of $L(s, \chi_0)$ at $s = 1$.

We suppose first that χ is a complex primitive character, and follow as closely as possible the argument of the preceding section.

Again the first term presents no difficulty; we have

$$-\frac{L'(\sigma, \chi_0)}{L(\sigma, \chi_0)} = \sum_1^\infty \chi_0(n)\Lambda(n)n^{-\sigma} \le -\frac{\zeta'(\sigma)}{\zeta(\sigma)} < \frac{1}{\sigma - 1} + c_1$$

for $1 < \sigma < 2$, where c_1 denotes a positive absolute constant (and similarly for $c_2, \ldots$ later[1]).

For the other two terms in (1), we have recourse to (17) of §12. This gives

$$-\Re\frac{L'(s, \chi)}{L(s, \chi)} = \tfrac{1}{2}\log\frac{q}{\pi} + \frac{1}{2}\Re\frac{\Gamma'(\frac{1}{2}s + \frac{1}{2}\mathfrak{a})}{\Gamma(\frac{1}{2}s + \frac{1}{2}\mathfrak{a})}$$
$$- \Re B(\chi) - \Re\sum_\rho\left(\frac{1}{s - \rho} + \frac{1}{\rho}\right),$$

where $\mathfrak{a}$ is 0 or 1. We can eliminate $B(\chi)$ and $\Sigma 1/\rho$ by appealing to (18) of §12. Since the Γ term above is $O[\log(t + 2)]$, we can express the result in the form

$$-\Re\frac{L'(s, \chi)}{L(s, \chi)} < c_2\mathscr{L} - \sum_\rho \Re\frac{1}{s - \rho}, \tag{2}$$

where we have written for brevity

$$\mathscr{L} = \log q + \log(t + 2). \tag{3}$$

This holds (for $\sigma > 1$) for any primitive χ, whether real or complex. Since

$$\Re\frac{1}{s - \rho} = \frac{\sigma - \beta}{|s - \rho|^2} \ge 0,$$

we can as before omit the series or any part of it.

We omit the whole series when estimating $L'(\sigma + 2it, \chi^2)/L(\sigma + 2it, \chi^2)$. There is the minor complication that χ^2, though nonprincipal, may not be primitive. However, if χ_1 is the primitive character that induces χ^2, it follows from (3) of §5 that

$$\left|\frac{L'(s, \chi^2)}{L(s, \chi^2)} - \frac{L'(s, \chi_1)}{L(s, \chi_1)}\right| \le \sum_{p|q}\frac{p^{-\sigma}\log p}{1 - p^{-\sigma}}$$
$$\le \sum_{p|q}\log p \le \log q.$$

Hence the upper bound in (2), namely $c_2\mathscr{L}$, remains valid.

[1] To leave the constants unnumbered, as we have done hitherto, would lead to confusion in the present section.

We choose t to be the ordinate γ of a zero $\beta + i\gamma$ of $L(s, \chi)$, and by retaining on the right of (2) only the one term corresponding to this zero, we obtain

$$-\Re \frac{L'(\sigma + it, \chi)}{L(\sigma + it, \chi)} < c_2 \mathscr{L} - \frac{1}{\sigma - \beta}.$$

The three estimates, when substituted in the basic inequality (1), give

$$\frac{4}{\sigma - \beta} < \frac{3}{\sigma - 1} + c_3 \mathscr{L}.$$

We take $\sigma = 1 + c_4/\mathscr{L}$, with a suitable c_4, and by the same argument as in the preceding section we obtain

$$\beta < 1 - c_5/\mathscr{L}. \tag{4}$$

This has been proved for any complex primitive χ, but the restriction to primitive χ can be removed, since, by (3) of §5, any zeros of $L(s, \chi)$ additional to those of $L(s, \chi_1)$, where χ_1 induces χ, are the zeros of a finite number of factors $1 - \chi_1(p)p^{-s}$ and are on $\sigma = 0$. We can accordingly assert that *there exists a positive absolute constant* c_5 *such that, if* χ *is a complex character to the modulus* q, *any zero* $\beta + i\gamma$ *of* $L(s, \chi)$ *satisfies* (4), *where*

$$\mathscr{L} = \log q + \log(|\gamma| + 2). \tag{5}$$

[We have modified the definition of $\mathscr{L}$ in (3) to accord with the choice $t = \gamma$.]

Suppose next that χ is a real primitive character. The preceding argument needs modification only in one respect: The inequality for $-\Re L'/L$ with $s = \sigma + 2it$ and χ replaced by χ^2 is no longer applicable since χ is the principal character. We must now relate L'/L to ζ'/ζ, and by the same argument as that used above when χ^2 was imprimitive, we have

$$\left| \frac{L'(s, \chi_0)}{L(s, \chi_0)} - \frac{\zeta'(s)}{\zeta(s)} \right| \leq \log q$$

for $\sigma > 1$. As regards $-\zeta'/\zeta$, we cannot quote the inequality (5) of §13, because this was proved only for large t. In proving it, a term $1/(s - 1)$ was neglected. When this is restored, the same argument as was used there gives

$$-\Re \frac{\zeta'(s)}{\zeta(s)} < \Re \frac{1}{s - 1} + c_6 \log (t + 2).$$

Hence

$$-\Re \frac{L'(\sigma + 2it, \chi^2)}{L(\sigma + 2it, \chi^2)} < \Re \frac{1}{\sigma - 1 + 2it} + c_7 \mathscr{L},$$

where $\mathscr{L}$ is again defined by (3).

Using this in place of the stronger inequality that was available for complex χ, we deduce from (1) that

$$\frac{4}{\sigma - \beta} < \frac{3}{\sigma - 1} + \Re \left(\frac{1}{\sigma - 1 + 2it}\right) + c_8 \mathscr{L},$$

where now $t = \gamma$. If we take $\sigma = 1 + \delta/\mathscr{L}$, and postulate that $\gamma \geq \delta/\mathscr{L}$, we get

$$\frac{4}{\sigma - \beta} < \frac{3\mathscr{L}}{\delta} + \frac{\mathscr{L}}{5\delta} + c_8 \mathscr{L},$$

whence

$$\beta < 1 - \frac{4 - 5c_8\delta}{16 + 5c_8\delta} \frac{\delta}{\mathscr{L}}.$$

If δ is sufficiently small in relation to c_8, we get an inequality of the form (4) but subject to the condition $\gamma \geq \delta/\mathscr{L}$, where $\mathscr{L}$ is given by (5). This condition is satisfied if $\gamma \geq \delta/\log q$. We have therefore proved that *there exists a positive absolute constant c_9 such that, if $0 < \delta < c_9$ and χ is a real nonprincipal character to the modulus q, then any zero $\beta + i\gamma$ of $L(s, \chi)$ for which*

$$|\gamma| \geq \frac{\delta}{\log q}$$

satisfies

$$\beta < 1 - \frac{\delta}{5\mathscr{L}},$$

where $\mathscr{L}$ is given by (5). We have omitted the requirement that χ should be primitive, for the same reason as before.

It remains to consider what can be proved about the zeros of $L(s, \chi)$, for real nonprincipal χ, with

$$|t| < \frac{\delta}{\log q},$$

where δ is a small positive constant. We shall show that there is at most one zero with $\sigma > 1 - \delta'/\log q$ for a suitable positive constant δ' and that, if there is one, it must be real. The final clause is in fact a corollary, for if there were a nonreal zero, there would be two zeros at conjugate complex points.

The inequality (2), with $s = \sigma > 1$, can be written

$$-\frac{L'(\sigma, \chi)}{L(\sigma, \chi)} < c_{10} \log q - \sum_{\rho} \frac{1}{\sigma - \rho},$$

the last sum being real since the zeros occur in conjugate complex pairs. In quoting this inequality we have assumed, as we may without loss of generality, that χ is primitive. If there were zeros at $\beta \pm i\gamma$, where $\gamma \neq 0$, we should have

$$-\frac{L'(\sigma, \chi)}{L(\sigma, \chi)} < c_{10} \log q - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2}.$$

For the left side, there is the crude lower bound

$$-\frac{L'(\sigma, \chi)}{L(\sigma, \chi)} = \sum_1^\infty \chi(n)\Lambda(n)n^{-\sigma} \geq -\sum_1^\infty \Lambda(n)n^{-\sigma} = \frac{\zeta'(\sigma)}{\zeta(\sigma)} > -\frac{1}{\sigma - 1} - c_{11}.$$

Thus

$$-\frac{1}{\sigma - 1} < c_{12} \log q - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2}.$$

We take $\sigma = 1 + 2\delta/\log q$; then

$$|\gamma| < \frac{\delta}{\log q} = \tfrac{1}{2}(\sigma - 1) < \tfrac{1}{2}(\sigma - \beta),$$

and the last inequality implies that

$$-\frac{1}{\sigma - 1} < c_{12} \log q - \frac{8}{5(\sigma - \beta)}.$$

If the δ of the previous result is sufficiently small in relation to c_{12}, we get $\beta < 1 - \delta/\log q$.

The argument is substantially the same if, instead of two conjugate complex zeros, there are two real zeros (or a double real zero). Thus we have proved: *There exists a positive absolute constant c_{13} such that, if $0 < \delta < c_{13}$, the only possible zero of $L(s, \chi)$ for a real nonprincipal χ, satisfying*

$$|\gamma| < \frac{\delta}{\log q}, \qquad \beta > 1 - \frac{\delta}{\log q}$$

is a single (simple) real zero.

The three results proved so far can be fitted together to give the following theorem, which we state for convenience of reference. It simplifies the statement to consider two cases according as $|t| \geq 1$ or $|t| < 1$, since in the former case the number $\mathscr{L}$ is essentially $\log q|t|$ and in the latter case it is essentially $\log q$.

THEOREM. *There exists a positive absolute constant c_{14} with the following property. If χ is a complex character modulo q, then $L(s, \chi)$ has no zero in the region defined by*

$$\sigma \geq \begin{cases} 1 - \dfrac{c_{14}}{\log q|t|} & \text{if } |t| \geq 1, \\[2ex] 1 - \dfrac{c_{14}}{\log q} & \text{if } |t| \leq 1. \end{cases} \tag{6}$$

If χ is a real nonprincipal character, the only possible zero of $L(s, \chi)$ in this region is a single (*simple*) *real zero.*

These results are due partly to Gronwall[2] and partly to Titchmarsh.[3]

We shall now prove a result due to Landau,[4] the effect of which is to assure us that if there exist values of q for which an L function formed with a real primitive character $(\bmod q)$ has a zero with $\beta > 1 - c/\log q$, then such values of q are very rare. He proved that *if χ_1, χ_2 are distinct real primitive characters to the moduli q_1, q_2 respectively, and if the corresponding L functions have real zeros β_1, β_2, then*

$$\min(\beta_1, \beta_2) < 1 - \frac{c_{15}}{\log q_1 q_2}, \tag{7}$$

where c_{15} is some positive absolute constant. The possibility that $q_1 = q_2$ is not excluded.

The proof is based on the fact that $\chi_1(n)\chi_2(n)$ is a character to the modulus $q_1 q_2$, being multiplicative and periodic. It is not in general a primitive character, but it is nonprincipal. For if $\chi_1(n)\chi_2(n) = 1$ whenever $(n, q_1 q_2) = 1$, we should have $\chi_1(n) = \chi_2(n)$ whenever $(n, q_1 q_2) = 1$, and this would mean that the primitive characters χ_1 and χ_2 would induce the same character to the modulus $q_1 q_2$. This is impossible by the results of §5.

[2] *Rendiconti di Palermo*, **35**, 145–159 (1913).

[3] *Rendiconti di Palermo*, **54**, 414–429 (1930); **57**, 478–479 (1933).

[4] *Göttinger Nachrichten*, **1918**, 285–295.

For $\sigma > 1$, we have

$$-\frac{L'(\sigma, \chi_1\chi_2)}{L(\sigma, \chi_1\chi_2)} < c_{16} \log q_1 q_2 ;$$

this is proved by the same argument as that which we applied to $L(s, \chi^2)$ when χ^2 was nonprincipal but not necessarily primitive. Further, by (2),

$$-\frac{L'(\sigma, \chi_j)}{L(\sigma, \chi_j)} < c_{17} \log q_j - \frac{1}{\sigma - \beta_j},$$

the symbol $\mathfrak{R}$ in (2) being now superfluous.

Now consider the expression

$$-\frac{\zeta'(\sigma)}{\zeta(\sigma)} - \frac{L'(\sigma, \chi_1)}{L(\sigma, \chi_1)} - \frac{L'(\sigma, \chi_2)}{L(\sigma, \chi_2)} - \frac{L'(\sigma, \chi_1\chi_2)}{L(\sigma, \chi_1\chi_2)}$$

$$= \sum_{n=1}^{\infty} \Lambda(n)[1 + \chi_1(n)][1 + \chi_2(n)]n^{-\sigma} \geq 0.$$

On substituting the previous upper bounds, and also that for $-\zeta'(\sigma)/\zeta(\sigma)$, we get

$$\frac{1}{\sigma - \beta_1} + \frac{1}{\sigma - \beta_2} < \frac{1}{\sigma - 1} + c_{18} \log q_1 q_2.$$

If σ is taken to be $1 + \delta/\log q_1 q_2$, for a sufficiently small positive constant δ, the last inequality shows that β_1 and β_2 cannot both be greater than $1 - \delta'/\log q_1 q_2$, for a suitable positive δ'. This proves (7).

Various deductions can be made from the last result. In particular one sees that *for at most one of the real nonprincipal characters χ (mod q) can $L(s, \chi)$ have a zero in the region* (6). [We assume here tacitly that c_{14}, in the definition (6), is diminished if necessary so as to satisfy $c_{14} \leq \frac{1}{2}c_{15}$.]

Another deduction concerns the possible sequence $q_1, q_2, \ldots$ of positive integers q with the property that there is a real primitive χ (mod q) for which $L(s, \chi)$ has a real zero β satisfying

$$\beta > 1 - c_{19}/\log q. \tag{8}$$

If c_{19} is suitably chosen, say $c_{19} = \frac{1}{3}c_{15}$, then

$$q_{j+1} > q_j^2.$$

For (7) implies that

$$1 - \frac{c_{19}}{\log q_j} < 1 - \frac{c_{15}}{\log q_j q_{j+1}},$$

whence the result.

A deduction made by Page[5] and applied by him to the prime number theorem for arithmetic progressions (see §20) concerns the set of positive integers $q \le z$, where $z \ge 3$. *If* c_{20} *is a suitable positive constant, there is at most one real primitive* χ *to a modulus* $q \le z$ *for which* $L(s, \chi)$ *has a real zero* β *satisfying*

$$(9) \qquad \beta > 1 - \frac{c_{20}}{\log z}.$$

The last inequality is, of course, of a somewhat more stringent nature than (8). The proof is immediate; if there were two such characters, both the zeros would satisfy

$$\beta > 1 - \frac{c_{20}}{\log z} \ge 1 - \frac{2c_{20}}{\log q_1 q_2},$$

and this would contradict (7) if $c_{20} = \frac{1}{2}c_{15}$.

If there is such an "exceptional" real character χ_1 to a modulus $q_1 \le z$, then q_1 will be a function of z, and the only real nonprincipal characters χ to moduli $q \le z$ for which $L(s, \chi)$ has a real zero satisfying (9) will be χ_1 and the imprimitive characters induced by χ_1. Their moduli will be multiples of q_1.

The only obvious general upper bound for a real zero β of an L function corresponding to a real primitive χ is that which can be derived from the lower bound for $L(1, \chi)$ provided by the class-number formula. Since $h(d) \ge 1$, the formulas (15) and (16) of §6, in which $d = \pm q$, give

$$(10) \qquad L(1, \chi) > c_{21} q^{-\frac{1}{2}}.$$

We can easily prove that

$$(11) \qquad |L'(\sigma, \chi)| < c_{22} \log^2 q \qquad \text{for } 1 - 1/\log q \le \sigma \le 1,$$

and it then follows, by the mean value theorem of the differential calculus, that

$$L(1, \chi) = L(1, \chi) - L(\beta, \chi) < (1 - \beta)c_{22} \log^2 q,$$

[5] *Proc. London Math. Soc.*, (2)**39,** 116–141 (1935).

whence

$$\beta < 1 - \frac{c_{23}}{q^{\frac{1}{2}}\log^2 q}. \tag{12}$$

By the usual argument, this holds also for real nonprincipal χ, even if imprimitive. If $\chi(-1) = 1$, which corresponds to the case $d > 0$, the inequality (12) can be improved[6] to the extent of a factor $\log q$, since in (16) of §6 there is a factor $\log \varepsilon$, and $\log \varepsilon \geq \log \frac{1}{2}(1 + q^{\frac{1}{2}})$.

The proof of (11) is as follows. We have

$$L'(\sigma, \chi) = -\sum_{1}^{\infty} \chi(n)(\log n)n^{-\sigma}$$

for $\sigma > 0$. Since $n^{-\sigma} = e^{-\sigma \log n} \leq en^{-1}$ for $n \leq q$, we have

$$\left| \sum_{n=1}^{q} \chi(n)(\log n)n^{-\sigma} \right| \leq e \sum_{n=1}^{q} (\log n)n^{-1} < c_{24} \log^2 q.$$

By partial summation, noting that $(\log n)n^{-\sigma}$ decreases for $n > q$, we have

$$\begin{aligned} \left| \sum_{n=q+1}^{\infty} \chi(n)(\log n)n^{-\sigma} \right| &\leq (\log q)q^{-\sigma} \max_N \left| \sum_{q+1}^{N} \chi(n) \right|. \\ &\leq (\log q)eq^{-1}q. \end{aligned}$$

These results imply (11).

We remark, for convenience of reference, that the same argument applied to $L(\sigma, \chi)$ gives

$$|L(\sigma, \chi)| < c_{25} \log q \qquad \text{for } 1 - 1/\log q \leq \sigma \leq 1. \tag{13}$$

In §21 we shall prove a theorem due to Siegel, which establishes a much more precise inequality for a real zero than that given in (12). But whereas all the results of the present section have been "effective," in the sense that numerical values could be assigned to all the constants, it does not seem to be possible to derive an effective inequality from Siegel's theorem.

[6] Goldfeld and Schinzel (*Ann. Scuola Norm. Sup. Pisa Cl. Sci.*, (4) **2**, 571–583 (1975)), have shown that $\beta < 1 - cq^{-1/2}$ if $\chi(-1) = -1$, and that $\beta < 1 - cq^{-1/2} \log q$ if $\chi(-1) = 1$.